WORKED SOLUTION

STEP II 2020, Question 5

Pure • Digit Sums and Divisibility

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

The key idea

Every power of 10 leaves remainder 1 on division by 9, so a number and its digit sum always agree modulo 9. That single fact turns each equation into a strong divisibility condition. A size argument then finishes the job: the digit sum grows only like $9n$ while the number itself grows like 10^{n-1}, so only a handful of candidates are left to test.

Solution

Part (i)

With $x = \sum_{r=0}^{n-1} a_r 10^r$ and $\mathrm{d}(x) = \sum_{r=0}^{n-1} a_r$,

$$x - \mathrm{d}(x) = \sum_{r=0}^{n-1} a_r \left(10^r - 1\right).$$

Each term has $a_r \geq 0$ and $10^r - 1 \geq 0$, so the sum is non-negative. Also $10^r - 1 = 99\ldots9$ (r nines) $= 9 \times 11\ldots1$, which is divisible by 9 (for $r = 0$ it is 0). So every term, and hence $x - \mathrm{d}(x)$, is divisible by 9. ■

In the language of congruences: $x \equiv \mathrm{d}(x) \pmod 9$ for every positive integer x. We use this repeatedly below.

Part (ii)

The equivalence. Since $\mathrm{d}(x) \equiv x \pmod 9$,

$$x - 44\,\mathrm{d}(x) \equiv x - 44x = -43x \equiv 2x \pmod 9,$$

because $-43 = -45 + 2$. So $x - 44\,\mathrm{d}(x)$ is a multiple of 9 exactly when $9 \mid 2x$. As $\gcd(2, 9) = 1$, $9 \mid 2x$ if and only if $9 \mid x$. This proves both directions at once. ■

The bounds. Suppose $x = 44\,\mathrm{d}(x)$ and x has n digits. Each digit is at most 9, so $\mathrm{d}(x) \leq 9n$ and $x = 44\,\mathrm{d}(x) \leq 396n$. And an n-digit number has leading digit $a_{n-1} \geq 1$,

so $x \geq 10^{n-1}$. Together,

$$10^{n-1} \leq x \leq 396n.$$

For $n = 5$ this would need $10\,000 \leq 1980$, which is false. Moreover, if $10^{n-1} > 396n$ for some $n \geq 5$ then $10^n = 10 \cdot 10^{n-1} > 3960n \geq 396(n+1)$, so by induction the inequality $10^{n-1} \leq 396n$ fails for every $n \geq 5$. Hence $n \leq 4$. ■

Ask yourself How can we cut the search down from all numbers up to 9999 to just a few?

Use the equivalence we have just proved. If $x = 44\,\mathrm{d}(x)$, then $x - 44\,\mathrm{d}(x) = 0$, which is a multiple of 9, so $9 \mid x$. Then $9 \mid 44\,\mathrm{d}(x)$, which forces $9 \mid \mathrm{d}(x)$ since $\gcd(44, 9) = 1$. Therefore $x = 44\,\mathrm{d}(x)$ is a multiple of $44 \times 9 = 396$.

Finding the solution. Since $n \leq 4$, we have $x \leq 396 \times 4 = 1584$, so x is one of 396, 792, 1188, 1584. Their digit sums are all 18, so $44\,\mathrm{d}(x) = 792$ in every case. The only one that works is

$$x = 792, \qquad \mathrm{d}(792) = 18, \quad 44 \times 18 = 792.$$

Every solution lies in our list of four candidates, and only 792 works, so there are no further values. ■

Part (iii)

Write $y = \mathrm{d}(x)$ and $z = \mathrm{d}(y)$, so that the equation is $x = 107z$, with $z = x/107 > 0$.

Divisibility. By part (i), $x \equiv y \equiv z \pmod 9$. So

$$0 = x - 107z \equiv z - 107z = -106z \pmod 9.$$

Now $106 = 99 + 7$, so $9 \mid 7z$, and since $\gcd(7, 9) = 1$ we get $9 \mid z$.

Size. Part (i) also gives $\mathrm{d}(m) \leq m$ for any positive integer m. So if x has n digits then $z = \mathrm{d}(y) \leq y \leq 9n$, and

$$10^{n-1} \leq x = 107z \leq 963n.$$

For $n = 5$ this needs $10\,000 \leq 4815$, which is false. The same induction as in part (ii) (now $10^n > 9630n \geq 963(n+1)$) shows that it fails for all $n \geq 5$, so $n \leq 4$.

Ask yourself The bound $z \leq 36$ still leaves several multiples of 9. Can we do better?

Yes, by applying the digit sum a second time. With $n \leq 4$, $y = \mathrm{d}(x) \leq 36$. Among $1, 2, \ldots, 36$ the largest digit sum is $\mathrm{d}(29) = 11$, because a two-digit number below 37 has tens digit at most 3, and tens digit 3 allows units digit at most 6 (digit sum at most 9), whereas tens digit 2 allows $2 + 9 = 11$. So $1 \leq z \leq 11$, and z is a multiple of 9. Hence $z = 9$ and

$$x = 107 \times 9 = 963.$$

Check: $\mathrm{d}(963) = 18$, $\mathrm{d}(18) = 9$ and $107 \times 9 = 963$. So $x = 963$ satisfies the equation. Any solution must have $z = 9$, and hence $x = 963$, so there are no further values. ■

Checks

- Spot-check part (i): $249 - 15 = 234 = 9 \times 26$.
- A computer search over every $x < 10^6$ finds 792 as the only solution of $x = 44\,\mathrm{d}(x)$ and 963 as the only solution of $x = 107\,\mathrm{d}(\mathrm{d}(x))$, in line with the bounds above.
- Notice $792 = 44 \times 18$ and $963 = 107 \times 9$: both are multiples of 9, as the divisibility arguments require.

Common slips

- Concluding only that x is a multiple of 9 in part (ii). The stronger fact that $\mathrm{d}(x)$ is also a multiple of 9, so that $396 \mid x$, is what makes the search short.
- Stating $n \leq 4$ after checking only $n = 5$. You need a reason why 10^{n-1} stays ahead of $396n$ for all larger n; the induction (or the observation that the left side grows by a factor of 10 while the right grows by a factor of at most $\frac{6}{5}$) supplies it.
- In part (iii), forgetting that $z > 0$, which is what rules out $z = 0$.

Beyond STEP

The fact that $x \equiv \mathrm{d}(x) \pmod 9$ is the theory behind "casting out nines", an old way of checking arithmetic. It is the first example of working in the ring $\mathbb{Z}/9\mathbb{Z}$, where 10 behaves like 1; the same idea with $10 \equiv -1$ gives the alternating-sum test for divisibility by 11. Equations mixing a number with functions of its digits are typically solved exactly as here: congruences cut down the possibilities and growth estimates make the list finite. Davenport's *The Higher Arithmetic* is a lovely introduction to congruences.

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